

The Reliability of the Linear-Method Precision Formula Under Spatial Heterogeneity

A Monte Carlo Study of Microfossil Concentration Estimation

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1 Declaration

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2 Abstract

Absolute microfossil concentration—the number of specimens per unit mass or volume of sediment—helps distinguish a genuine decline in abundance from an apparent decline in percentage caused by increases in other groups. It is commonly estimated by adding a known dose of marker grains and counting targets and markers together. The uncertainty of this estimate is usually calculated using a precision formula that assumes specimens are randomly and uniformly distributed. This report tested whether formula-based precision still agreed with the variation observed across repeated simulations when specimen locations were non-uniform. We compared a random uniform reference pattern with three controlled non-uniform patterns across four target-to-marker ratios and two stopping counts, giving 32 settings. Each setting included 400 repetitions, with targets and markers sharing the same spatial distribution.

Both the formula-based and simulation-based estimates of uncertainty increased as the target-to-marker ratio increased. Their mean relative difference was smallest at a ratio of 2 (2.0%) and largest at a ratio of 10 (7.9%). A stopping count of 800 produced a smaller difference than a stopping count of 333 in 12 of the 16 matched comparisons. The centre-high pattern differed most strongly from the random uniform pattern, but it produced the largest precision difference in only one of the eight combinations of ratio and stopping count. The difference also tended to decrease when more markers were counted. However, marker count changed together with ratio

and stopping count, so their separate effects could not be determined. The concentration estimate based on the mean marker count had an absolute relative bias ranging from 0.02% to 1.52%, with a mean of 0.45%. Within the settings examined, stronger non-uniformity was therefore not consistently linked to poorer agreement between the two precision estimates. This is an initial result for cases in which targets and markers share the same spatial distribution, rather than a general validation of the precision formula.

3 Introduction

Microfossils preserved in sediment provide evidence of past ecological and environmental change. Interpreting this evidence requires knowing how abundant a microfossil group actually was, not only how common it was relative to other groups. A group's percentage can fall because another group becomes more abundant, even when its own abundance does not change. Absolute concentration, defined as the number of target specimens per unit mass or volume of sediment, helps distinguish genuine changes in a group from changes caused by the composition of other microfossil groups (Mays, Amores, and Mays 2025, 2–3).

Because counting every specimen is impractical, concentration is commonly estimated by adding a known dose of exotic marker grains and counting targets and markers together until a preset number of targets is reached. The expected uncertainty of this estimate is summarised by a precision formula. Its counting terms assume that targets and markers are independently and uniformly distributed, but real microscope preparations need not satisfy this assumption. Mays et al. note that the effects of inhomogeneity had not been comprehensively assessed because suitable quantitative data were not yet available (Mays, Amores, and Mays 2025, 28).

This report asks whether formula-based precision continues to agree with the variation observed across repeated simulations when specimen locations are non-uniform. Formula-based precision is compared with a simulation benchmark under a random uniform reference pattern and three non-uniform patterns, across four target-to-marker ratios and two counting efforts. Within each setting, targets and markers follow the same spatial distribution. The contribution is therefore a controlled quantitative comparison for cases in which both groups become non-uniform in the same way; cases in which they follow different spatial distributions remain outside the present scope.

4 Background

4.1 Exotic-marker estimation and the linear method

Absolute concentration is the number of target specimens per unit mass or volume of sediment. It provides information that relative abundance alone cannot provide. For example, the percentage of one group may change because that group changes, because other groups change, or because both change. Absolute concentration helps distinguish between these possibilities (Mays, Amores, and Mays 2025, 2–3). It is therefore useful when biological abundance must be compared between samples or time periods.

The exotic-marker technique estimates absolute concentration by adding a known quantity of marker specimens to a sample. Benninghoff (1962) introduced this approach to microfossil analy-

sis using a measured suspension of exotic pollen. Stockmarr (1971, 616–19) later described soluble tablets containing *Lycopodium* spores as a convenient way to add marker specimens. As summarised by Mays et al. (2025, 2–3), targets and markers are counted together, and their count ratio is used to estimate the concentration of targets in the original sample.

The method assumes that targets and markers are sufficiently well mixed for the counted area to represent their ratio in the preparation. Ideally, the two specimen types should have similar physical and chemical behaviour, reducing the chance that one type is overrepresented in the counted material (Mays, Amores, and Mays 2025, 2–3).

Maher (1981, 156) describes the conventional procedure in which an analyst counts until a preset number of target specimens has been reached while also recording every marker encountered. Mays et al. (2025, 4) refer to this procedure as the linear method. In this report, the preset target count is denoted by x_{stop} and is called the stopping count. Counting was carried out within a horizontal transect band; the transect geometry and stopping procedure are specified in the Methods section.

Mays et al. (2025, Eq. 1, p. 3) present the general concentration equation as following Benninghoff, with its terms updated from Maher. Using their notation, N_1 is the number of marker doses added to the sample and $\bar{Y}_1$ is the mean number of markers per dose. The nominal total number of markers added is therefore

$$\bar{M} = N_1 \bar{Y}_1. \quad (1)$$

Let x be the number of targets counted, n the number of markers counted, and V the original sample volume. Scaling the counted target-to-marker ratio by the nominal marker total gives the concentration estimate

$$\hat{c} = \frac{x\bar{M}}{nV}, \quad (2)$$

where $\hat{c}$ is the estimated number of targets per unit volume.

In this report, $\bar{M}$ is the nominal total for the five-tablet marker dose, rather than the exact number of markers in an individual preparation. Stockmarr’s measurements of five-tablet groups showed that the total spore content was not identical between replicate groups; the estimated standard error was approximately 3.0% of the mean (1971, 618–19). The uncertainty in marker dose is therefore included in the formula-based precision calculation introduced in the next section.

Laboratory preparation can introduce additional uncertainty, but it is not simulated here. This report begins with generated target and marker locations and focuses on uncertainty arising during counting. Evidence concerning marker loss during laboratory preparation is summarised in Appendix Section 10.1.1.

4.1.1 Averaging repeated concentration estimates

In this project, the target count x is fixed by the stopping rule, but the number of counted markers n changes between repetitions. Because n appears in the denominator of Equation (2), repetitions with small marker counts can have a relatively large effect on the average concentration estimate.

For positive marker counts, $1/n$ is a convex function, so Jensen's inequality gives

$$E(1/n) \geq 1/E(n),$$

where E denotes the expected value over the random marker count. Therefore, directly averaging the individual concentration estimates can give a higher value than calculating the concentration from the mean marker count. For this reason, the project also uses an estimate based on the mean marker count. Its calculation is described in the Methods section.

4.2 Formula-based precision and the target-to-marker ratio

Stockmarr (1971, 620) introduced an approximate total-error calculation for exotic-marker concentration estimates. Mays et al. (2025, Eq. 2, pp. 4–6) present this calculation in the notation used for the linear method, using terms defined by Maher (1981). The formula is

$$\sigma_L = 100 \sqrt{\left(\frac{s_{1P}}{\sqrt{N_1}}\right)^2 + \left(\frac{\sqrt{x}}{x}\right)^2 + \left(\frac{\sqrt{n}}{n}\right)^2}, \quad (3)$$

Here, σ_L is the proportional standard error of the concentration estimate, reported as a percentage; equivalently, it is the coefficient of variation multiplied by 100. The quantity s_{1P} is the relative standard deviation of the marker count in one dose, N_1 is the number of doses added, and x and n are the counted target and marker numbers. The three terms represent uncertainty from the marker dose, target count and marker count.

The target- and marker-counting terms follow a Poisson model, under which their squared relative contributions reduce to $1/x$ and $1/n$ (Mays, Amores, and Mays 2025, 6). Although x is fixed by the stopping rule, Mays et al. explain that its contribution remains because the area examined before reaching x varies between repeated counts.

For this project, the calibrated five-tablet group reported by Stockmarr (1971, 618–19) was treated as one complete marker dose. Therefore, $N_1 = 1$, $\sqrt{N_1} = 1$, and the first term in Equation (3) reduces to $s_{1P,5}^2$. This is why N_1 does not appear in the project-specific formula in the Methods section. The numerical value of $s_{1P,5}$ is specified there.

4.2.1 Target-to-marker ratio

Maher (1981, 156) uses u for the underlying ratio of targets to markers. In this report, the fixed simulation setting is

$$\bar{u} = \frac{N_T}{\bar{M}},$$

where N_T is the target total and $\bar{M}$ is the nominal marker total. The bar distinguishes this setting from the realised ratio, which varies because the realised marker total changes between repetitions.

Regal and Cushing (1979, Eq. 6.2, p. 564) showed that, under a large-sample Poisson model and fixed expected counting effort, concentration variance is minimised when target and marker

concentrations are equal. This corresponds to $u = 1$. A short mathematical explanation is provided in Appendix Section 10.1.2.

Maximum theoretical precision is not identical to maximum practical efficiency. Maher (1981, 156–57) noted that at $u = 1$, half of the counted specimens are markers rather than biological targets. Mays et al. (2025, 4) recover the equal-abundance precision result and identify a ratio close to 2 : 1 as an effective balance between precision and counting effort when combined counts are approximately 500 or greater. Price et al. (2016, 78) recommend ratios below 5 : 1, and ideally between 0.5 : 1 and 2 : 1, in their dinoflagellate-cyst experiments.

Accordingly, $\bar{u} = 1$ represents the theoretical equal-abundance case and $\bar{u} = 2$ represents the practical value discussed by Maher and Mays et al. The values $\bar{u} = 5$ and 10 examine conditions with progressively fewer markers. These settings are not intended to determine a universal optimal ratio.

4.3 Spatial randomness, heterogeneity and the CSR diagnostic

Complete spatial randomness (CSR) provides the uniform comparison case for this project. Under CSR, points are placed independently and every location in the study region has the same chance of containing a point (Diggle 2013). When the total number of points is fixed, this can be produced by drawing the horizontal and vertical coordinates independently from a Uniform(0, 1) distribution. The resulting pattern is random, but has no systematic concentration towards any part of the unit square.

Real preparations may not follow this uniform pattern. Specimens can be more concentrated in the centre or towards one side of a slide. The Beta distribution provides a simple way to represent these patterns on the interval [0, 1]. A symmetric Beta distribution with equal parameters greater than one places more points near the centre. Unequal parameters can instead place more points towards the right or left. Applying these distributions to both coordinates produces a centre-high pattern, while applying them only to the horizontal coordinate produces right-high and left-high patterns.

These patterns were selected to represent clear departures from CSR rather than to reproduce a particular microscope slide. Within each spatial scenario, targets and markers followed the same coordinate distributions. This represents a preparation in which both groups share the same broad spatial pattern, while their observed counts can still differ by chance. The exact Beta parameters used in the final simulation are given in the Methods section.

4.3.1 Checking departure from CSR

Departure from CSR was assessed using inter-event distances (Diggle 2013, Eq. 2.2, pp. 18–19). The empirical inter-event distance distribution was compared with the corresponding theoretical CSR distribution for the unit square. A larger maximum difference between the two curves indicates a stronger departure from CSR. The mathematical definition is given in Appendix Section 10.1.3, and its implementation is described in the Methods section.

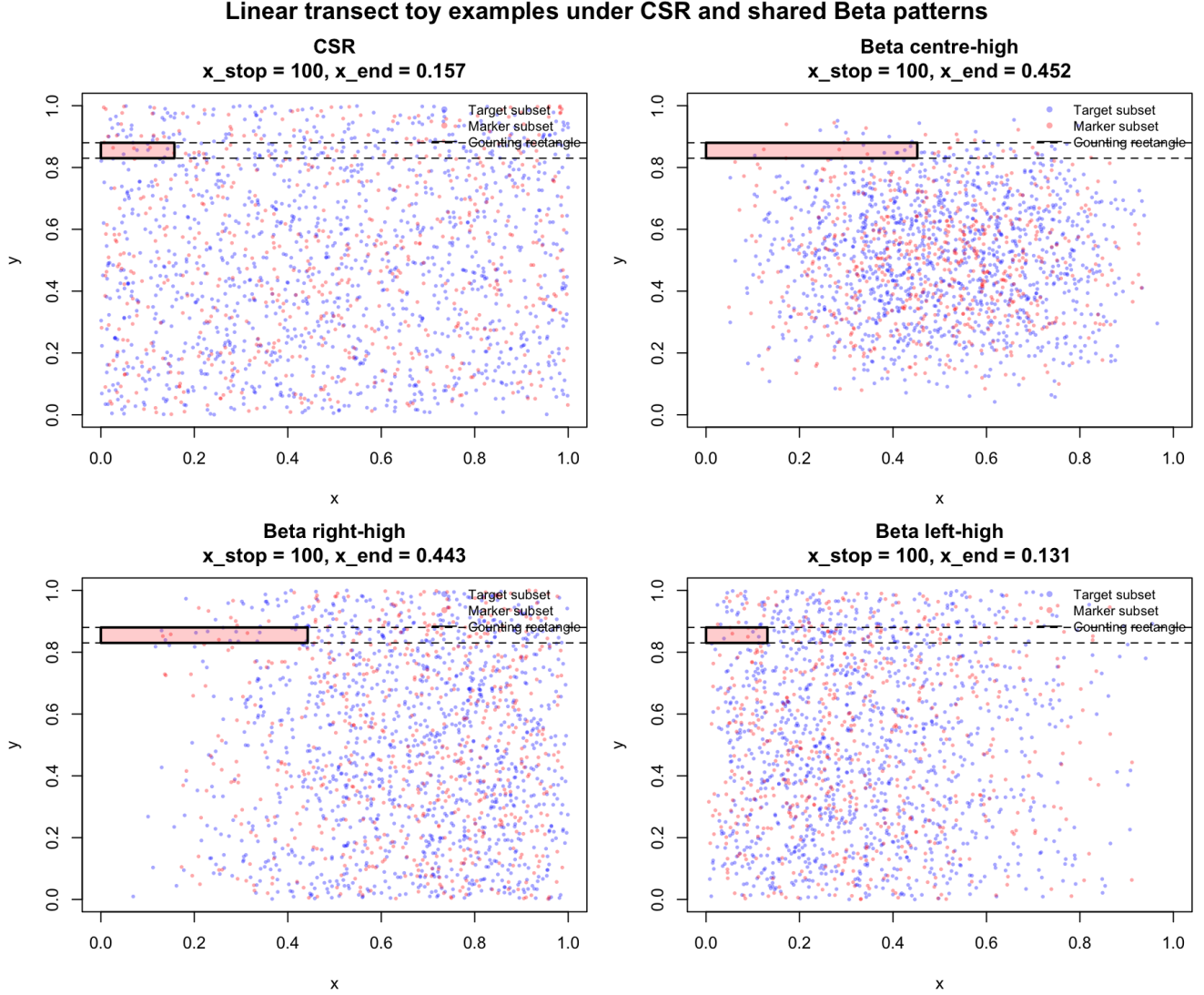


Figure 1: Toy illustration of target and marker locations under CSR and three non-uniform Beta patterns, together with the horizontal transect band and the final counting rectangle. A stopping count of $x_{\text{stop}} = 100$, reduced point totals and simplified Beta settings are used here so that the counting process can be seen clearly. The exact parameter values and stopping counts used in the final simulation are given in the Methods section.

4.4 Previous research and the present question

The studies reviewed above establish the exotic-marker concentration estimator, the linear counting method and its formula-based precision calculation. Mays et al. (2025, 6) explain that the target- and marker-counting terms are based on independent Poisson distributions. They also note that specimens on microscope slides need not follow random Poisson patterns and that the effects of inhomogeneity had not been comprehensively assessed because suitable quantitative data were not yet available (Mays, Amores, and Mays 2025, 28).

This provides the motivation for the present project. The project asks whether the formula-based precision estimate continues to agree with the variation observed across repeated simulations when

specimen locations depart from CSR. It compares formula-based and simulation-based precision under CSR and three non-uniform Beta patterns, across four target-to-marker ratios and two stopping counts. Within each scenario, targets and markers follow the same coordinate distributions.

5 Methods

5.1 Overview of the simulation

This was a Monte Carlo study of simulated microscope-slide patterns rather than observed data. Because all target and marker locations were generated directly, no missing-data treatment or preprocessing was required.

The project compared formula-based precision with a simulation benchmark across four spatial patterns, four target-to-marker ratios and two stopping counts. Figure 2 summarises the procedure. Exact settings and calculations are given in the following subsections.

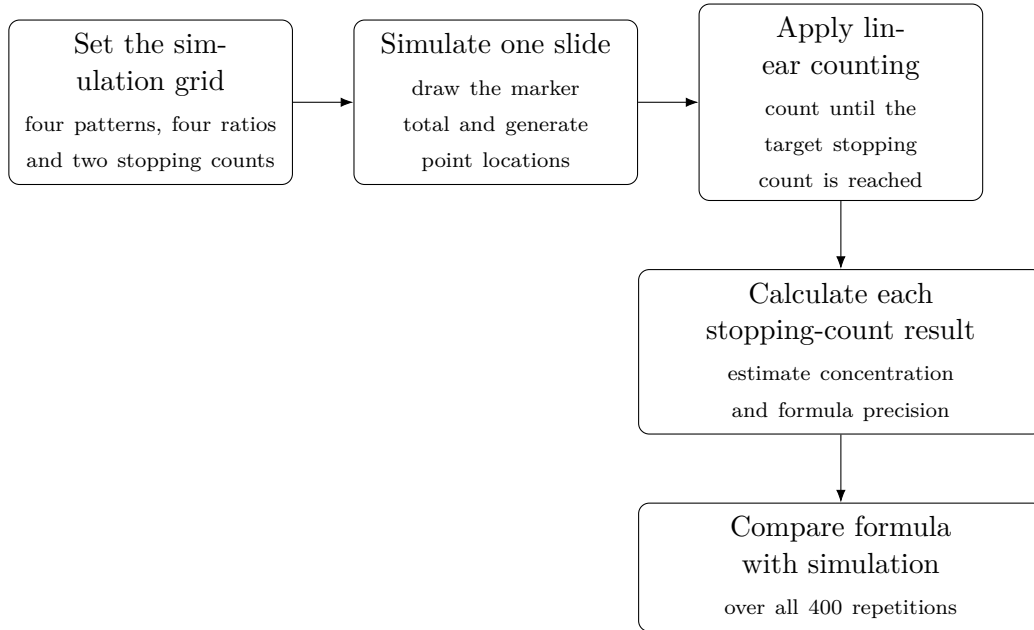


Figure 2: Workflow of the formal Monte Carlo simulation. For each combination of spatial pattern, target-to-marker ratio and stopping count, the formula-based precision estimate was compared with the simulation benchmark calculated from 400 repetitions.

The complete workflow is implemented in `01_run_final_simulation.R`. The formal settings are specified on lines 31 to 140, and coordinates are generated by `generate_xy()` on lines 149 to 184. Each combination of spatial pattern and target-to-marker ratio is processed by `run_one_setting()` on lines 202 to 381. The raw results are combined and written on lines 515 to 554, and the summaries are calculated and written on lines 561 to 745.

5.2 Simulation design and spatial settings

The simulation varied three settings: spatial pattern, target-to-marker ratio and target stopping count. The target-to-marker ratio $\bar{u}$ is defined in Background Subsection 4.2.1. Four values, $\bar{u} \in \{1, 2, 5, 10\}$, were used for the reasons summarised there.

With the nominal marker total fixed at $\bar{M} = 105000$, these ratios gave target totals of 105000, 210000, 525000 and 1050000, respectively. The realised marker total varied between repetitions, while $\bar{u}$ remained a fixed simulation setting.

Each target and marker was represented by a horizontal coordinate X and a vertical coordinate Y in the unit square. The coordinate distributions used for each spatial pattern are shown in Table 1.

Spatial pattern	X distribution	Y distribution
CSR	Uniform(0, 1)	Uniform(0, 1)
Centre-high	Beta(4, 4)	Beta(4, 4)
Right-high	Beta(5, 1.5)	Uniform(0, 1)
Left-high	Beta(1.5, 5)	Uniform(0, 1)

Table 1: Coordinate distributions used for the four spatial patterns. Targets and markers were generated independently but followed the same coordinate distributions within each pattern.

The Beta parameters were chosen to create clear departures from CSR while leaving enough target points in the transect band for both stopping counts to be examined. They were controlled simulation settings rather than estimates from observed microscope slides. The right-high pattern placed more points near the right edge, while the left-high pattern placed more points near the left edge.

Using the same spatial pattern for targets and markers kept their expected ratio constant across the unit square. This allowed non-uniform spatial patterns to be studied while targets and markers remained distributed in the same way. Situations in which targets and markers follow different spatial distributions were outside the scope of this project.

The complete parameter settings are listed in Table 2.

Stockmarr reported a mean of 105022 marker spores and a standard deviation of 3112 for a five-tablet dose (1971, 618–19). The mean was rounded to the nominal value $\bar{M} = 105000$, which Stockmarr also used in the concentration calculation. At the beginning of each repetition, the realised marker total was drawn from a Normal distribution with mean 105000 and standard deviation 3112. It was then rounded to an integer and given a lower bound of one.

The full Stockmarr marker total was used rather than a smaller total chosen only to reduce computation. This required more simulated points, but kept the marker-dose variation in the simulation consistent with the value used in the precision formula.

The point-generation model did not reproduce the independent Poisson counting assumption exactly. The target total was fixed, and the realised marker total was fixed within each repetition after it had been drawn. Consequently, for any fixed region, the spatial counts were binomial conditional on their totals rather than unconditionally Poisson. The final counting rectangle was also random because its endpoint was determined by a target order statistic. The Poisson terms in Equation (11) were nevertheless retained because they are part of the precision formula being

Setting	Value
Nominal marker total $\bar{M}$	105000
Realised marker total	Normal mean 105000, SD 3112; rounded, minimum 1
Sample volume V	1
Spatial domain	Unit square $[0, 1]^2$
Transect band	$y \in [0.83, 0.88]$
Target-to-marker ratio $\bar{u}$	1, 2, 5, 10
Target total N_T	105000, 210000, 525000, 1050000
Stopping count x_{stop}	333, 800
Planned repetitions per setting M	400
Main seed	123 with a fixed offset for each setting

Table 2: Parameter values used in the formal simulation. A setting was one combination of spatial pattern, target-to-marker ratio and stopping count.

evaluated (Mays, Amores, and Mays 2025, 6). The difference between the conditional binomial and Poisson models was expected to be small here: the largest designed target sampling fraction was $800/105000 = 0.76\%$, and the expected marker sampling fractions were similarly small. The results should therefore be interpreted as a test of the formula under a conditional finite-population point process, rather than under its exact Poisson data-generating assumptions.

The stopping count $x_{\text{stop}} = 333$ gives an expected marker count of approximately 166 when $\bar{u} = 2$, giving an expected combined count close to 500. The larger stopping count, $x_{\text{stop}} = 800$, was included to examine the same settings with greater counting effort.

For each combination of spatial pattern and target-to-marker ratio, 400 independent point patterns were generated. Each pattern was evaluated at both stopping counts. The design therefore contained $4 \times 4 \times 400 = 6400$ simulated point patterns and 12800 planned stopping-count records across 32 settings.

The main seed was 123. For spatial-pattern index $i = 1, \dots, 4$ and ratio index $j = 1, \dots, 4$, the seed for each setting was

$$123 + 100000i + 1000j. \quad (4)$$

A stopping-count result was retained only if the transect band contained at least x_{stop} targets and the final counting rectangle contained at least one marker. The number of valid results retained from the planned $M = 400$ repetitions is denoted by M_{eff} . These conditions were fixed before any results were summarised.

The parameter values and coordinate distributions are implemented in `01_run_final_simulation.R` on lines 31 to 184. The realised marker total and point locations are generated on lines 256 to 271, the validity conditions are applied on lines 315 to 337, and the seed for each setting is calculated on lines 388 to 416. The script writes the settings to `outputs/full_R400/simulation_parameters.csv` and `outputs/full_R400/scenario_parameters.csv`.

5.3 CSR diagnostic using inter-event distances

This diagnostic was run separately from the main Monte Carlo simulation to check whether the four generated spatial patterns behaved as intended. For each spatial scenario $i = 1, \dots, 4$, one representative target pattern was generated at $\bar{u} = 2$, giving $N_T = 210000$. The seed used to generate each pattern was $987 + 1000i$. A random subset of 1000 target points was then selected using the seed $987 + 30i$. Only target locations were used in this diagnostic.

All inter-event distances among the selected points were calculated. Their empirical cumulative distribution $\hat{H}(t)$ was evaluated at 500 equally spaced values between 0 and 1 and compared with the theoretical CSR curve $H_{\text{CSR}}(t)$ in Equation (18). The CSR gap was defined as

$$\text{CSR gap} = \max_{0 \leq t \leq 1} |\hat{H}(t) - H_{\text{CSR}}(t)|. \quad (5)$$

The CSR gap summarises the largest difference between the two curves. It is a descriptive measure rather than a p -value, and it is separate from the precision gap defined later.

The representative patterns are generated in `03_make_spatial_diagnostics.R` on lines 135 to 157. The theoretical CSR curve is implemented on lines 484 to 497, and the diagnostic is calculated by `draw_csr_diagnostic()` on lines 513 to 641. The script writes `outputs/spatial_diagnostics/csr_gap_summary.csv` and saves the diagnostic figure as `fig_csr_diagnostic_inter_event.png` and `fig_csr_diagnostic_inter_event.pdf` in the same folder.

5.4 Linear-transect counting algorithm

Each point pattern generated using the design in Subsection 5.2 was processed using the following linear counting algorithm. Figure 1 shows the transect band and final counting rectangle under the four spatial patterns using toy examples with $x_{\text{stop}} = 100$ and simplified parameters.

1. Retain the target and marker points within the horizontal transect band $y \in [0.83, 0.88]$.
2. Order the horizontal coordinates of the target points in the band.
3. For each stopping count, define x_{end} as the horizontal coordinate of the x_{stop} -th target.
4. Define the final counting rectangle as $[0, x_{\text{end}}] \times [0.83, 0.88]$.
5. Record the target count as $x = x_{\text{stop}}$ and record the number of markers in the rectangle as n .

The procedure was applied separately at $x_{\text{stop}} = 333$ and $x_{\text{stop}} = 800$ using the same generated point pattern. The validity rules defined in Subsection 5.2 were then applied. A result was discarded if the band contained fewer than x_{stop} targets or if the final rectangle contained no markers.

The algorithm is implemented in `run_one_setting()` in `01_run_final_simulation.R`. Lines 273 to 305 select the points in the transect band and order the target coordinates. Lines 307 to 337 apply the stopping counts, record x_{end} , count the markers and apply the validity rules.

5.5 Concentration and precision calculations

For each setting, the calculations used the valid repetitions $m = 1, \dots, M_{\text{eff}}$, where M_{eff} is defined in Subsection 5.2.

5.5.1 Concentration estimates

The concentration estimate for repetition m was

$$\hat{c}_m = \frac{x_m \bar{M}}{n_m V}, \quad (6)$$

where $x_m = x_{\text{stop}}$ is the counted target number and n_m is the counted marker number. The estimator used the nominal marker total $\bar{M} = 105000$ and sample volume $V = 1$. The repetition-specific realised marker total was used only to generate the marker locations.

The known true concentration was

$$c_{\text{true}} = \frac{N_T}{V} = \frac{\bar{u} \bar{M}}{V}. \quad (7)$$

Since $V = 1$, the true concentration was equal to the fixed target total N_T . For each setting, the mean counted marker number was

$$\bar{n} = \frac{1}{M_{\text{eff}}} \sum_{m=1}^{M_{\text{eff}}} n_m. \quad (8)$$

Jensen's inequality in Subsection 4.1.1 shows that averaging the individual estimates $\hat{c}_m$ can differ from calculating concentration from the mean marker count. Therefore, a second estimate based on $\bar{n}$, denoted by $\hat{c}_{\text{Jensen}}$, was calculated as

$$\hat{c}_{\text{Jensen}} = \frac{x_{\text{stop}} \bar{M}}{\bar{n} V}. \quad (9)$$

Its agreement with the truth was summarised by the absolute relative bias

$$b = \frac{|\hat{c}_{\text{Jensen}} - c_{\text{true}}|}{c_{\text{true}}}. \quad (10)$$

5.5.2 Formula-based precision estimate

Using the five-tablet dose convention described in Background Section 4.2, Equation (3) was implemented for repetition m as

$$\sigma_{L,m} = 100 \sqrt{s_{1P,5}^2 + \frac{1}{x_m} + \frac{1}{n_m}}, \quad (11)$$

where the relative standard deviation of the five-tablet marker dose was

$$s_{1P,5} = \frac{3112}{105000}. \quad (12)$$

The mean formula-based precision estimate for each setting was

$$\bar{\sigma}_L = \frac{1}{M_{\text{eff}}} \sum_{m=1}^{M_{\text{eff}}} \sigma_{L,m}. \quad (13)$$

5.5.3 Simulation-based precision benchmark

The simulation-based precision benchmark was calculated from the squared relative deviations of the individual concentration estimates from the known true concentration:

$$\sigma_{\text{exact}} = \frac{100}{c_{\text{true}}} \sqrt{\frac{\sum_{m=1}^{M_{\text{eff}}} (\hat{c}_m - c_{\text{true}})^2}{M_{\text{eff}} - 1}}. \quad (14)$$

The denominator $M_{\text{eff}} - 1$ follows the calculation used by Mays et al. (2025, Eq. S18) and keeps the benchmark comparable with their simulation results. When $M_{\text{eff}} = 400$, using this denominator rather than M_{eff} increases σ_{exact} by about 0.125%. No additional corrections were applied.

5.5.4 Precision-gap measures

The formula-based estimate and simulation benchmark were compared using signed, absolute and relative precision gaps:

$$\Delta = \bar{\sigma}_L - \sigma_{\text{exact}}, \quad (15)$$

$$|\Delta| = |\bar{\sigma}_L - \sigma_{\text{exact}}|, \quad (16)$$

$$\text{Relative precision gap} = \frac{|\Delta|}{\sigma_{\text{exact}}}. \quad (17)$$

A negative Δ means that the formula-based estimate was below the simulation benchmark, while a positive value means it was above. The relative gap allows settings with different values of σ_{exact} to be compared on the same scale.

The true concentration is calculated in `01_run_final_simulation.R` on lines 211 to 215. The individual concentration and formula-based precision estimates are calculated on lines 339 to 353. The mean marker count, concentration bias, simulation benchmark and precision gaps are calculated by `summarise_group()` on lines 630 to 665.

5.6 Relationship between precision gap and mean counted marker number

To examine whether the relative precision gap was related to the number of markers actually counted, it was plotted against the mean counted marker number $\bar{n}$ for all 32 settings. The horizontal axis used a logarithmic scale, the four spatial patterns were shown by colour, and the two stopping counts were shown by plotting symbol. This was a descriptive comparison and did not treat $\bar{n}$ as a cause of the precision gap.

The plot is produced by `draw_gap_vs_nbar()` in `02_make_report_figures.R` on lines 372 to 468 and saved on lines 470 to 477 as `fig_gap_vs_nbar_pooled.png` and `fig_gap_vs_nbar_pooled.pdf`.

5.7 Software, outputs and reproducibility

All simulations and analyses were carried out in R version 4.4.2. The code and outputs required to reproduce the report are available at https://github.com/SHERRY-DOGGIE-1/MDS_Final_Report. Table 3 summarises the four scripts used in the analysis.

Script	Role and main files
<code>01_run_final_simulation.R</code>	Runs the formal simulation and writes the raw and summary results (<code>all_raw_results.csv</code> and <code>summary_results.csv</code>), together with supporting parameter, checkpoint and timing files, to <code>outputs/full_R400/</code> .
<code>02_make_report_figures.R</code>	Reads <code>outputs/full_R400/summary_results.csv</code> and writes the five result figures and <code>table_main_results.csv</code> to <code>outputs/full_R400/report_figures/</code> .
<code>03_make_spatial_diagnostics.R</code>	Generates the spatial examples and CSR diagnostic, and writes the diagnostic summaries and figures to <code>outputs/spatial_diagnostics/</code> .
<code>wk2_toy_beta.R</code>	Generates Figure 1 and writes <code>fig4_linear_transect_examples.png</code> to <code>partB_beta_toy_figures/</code> .

Table 3: Scripts used to generate the formal simulation results, spatial diagnostics and Background illustration.

`01_run_final_simulation.R` was run before `02_make_report_figures.R`. The spatial-diagnostic and toy-figure scripts produced independent outputs. Exact seeds and code locations are given in the relevant Methods subsections.

Detailed line references for `01_run_final_simulation.R` and `03_make_spatial_diagnostics.R` are given in the preceding subsections. `02_make_report_figures.R` reads the saved summary on lines 58 to 61, produces the result figures on lines 285 to 477, and writes the report table on lines 498 to 510. Figure 1 is produced by `wk2_toy_beta.R` on lines 327 to 351.

6 Results

6.1 Simulation checks

All 32 reported settings retained all 400 planned repetitions, so $M_{\text{eff}} = 400$ and the success rate was 1.00 throughout. No repetition failed to reach the target stopping count or contained zero markers in the final counting rectangle. Across the 6400 generated point patterns, the realised marker total ranged from 94516 to 116118, with an overall mean of approximately 104961. Across the 16 scenario–ratio combinations, the mean realised marker total ranged from 104617 to 105156.

6.2 CSR diagnostic

Figure 3 compares the empirical inter-event distance curve with the theoretical CSR curve for each spatial scenario, using the diagnostic described in Subsection 5.3. The CSR scenario had the smallest gap, 0.004. The gaps were 0.192 for right-high, 0.203 for left-high and 0.420 for centre-high.

6.3 Formula versus simulation precision

Figure 4 shows the relative precision gap for all 32 settings, and Table 4 reports the formula-based and simulation precision estimates at $\bar{u} = 10$. Complete values for all settings are reported in Appendix Table 5.

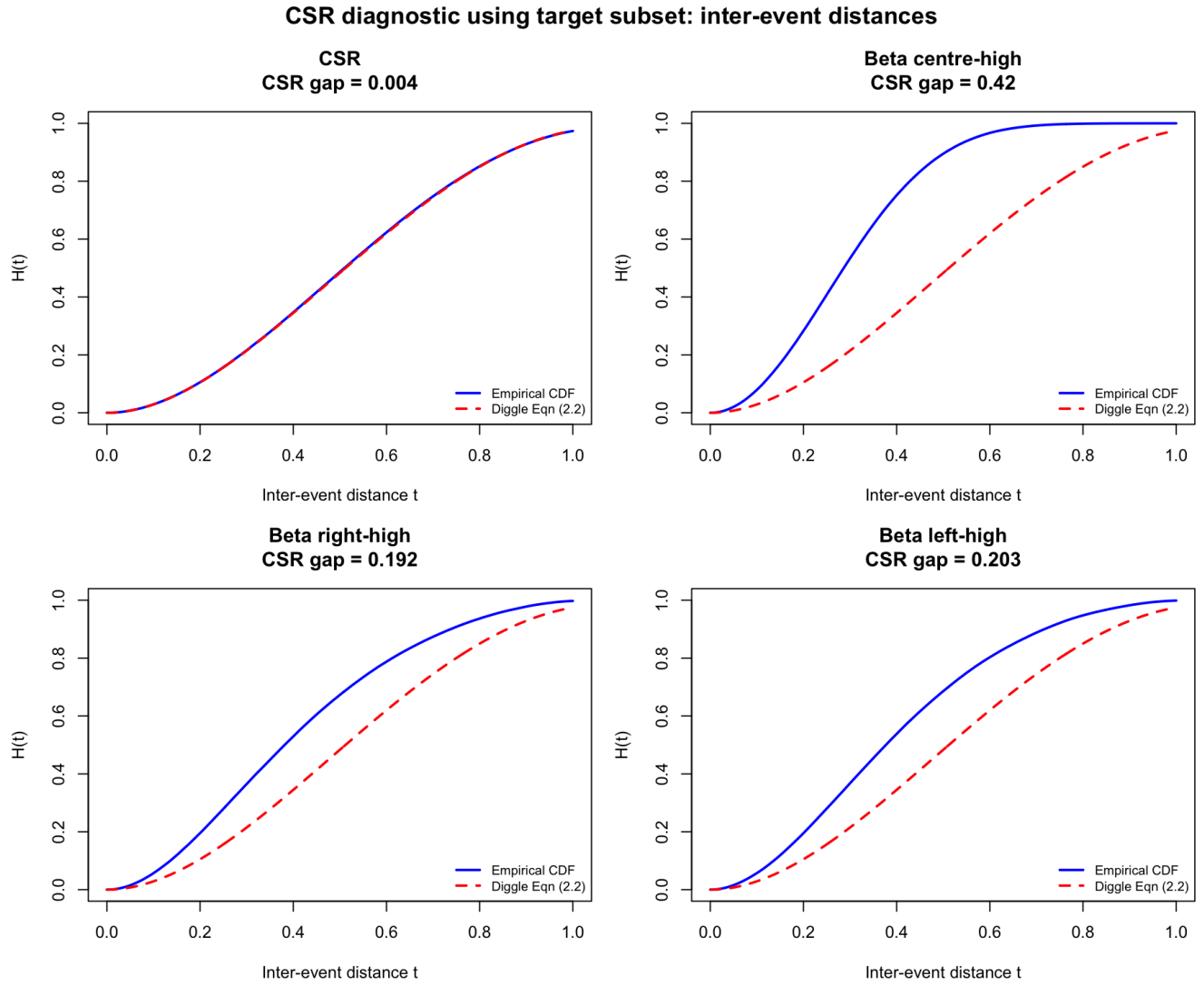


Figure 3: Empirical inter-event distance curves (solid) and the theoretical CSR curve (dashed) for the four spatial scenarios.

Table 4: Formula-based and simulation precision estimates at $\bar{u} = 10$ for both stopping counts.

Scenario	x_{stop}	$\bar{\sigma}_L$ (%)	σ_{exact} (%)	Rel. gap
Beta centre-high	333	18.62	20.95	0.111
Beta left-high	333	18.77	21.59	0.131
Beta right-high	333	18.57	20.52	0.095
CSR	333	18.55	20.12	0.078
Beta centre-high	800	12.15	12.83	0.053
Beta left-high	800	12.18	12.88	0.055
Beta right-high	800	12.15	12.51	0.029
CSR	800	12.12	13.22	0.083

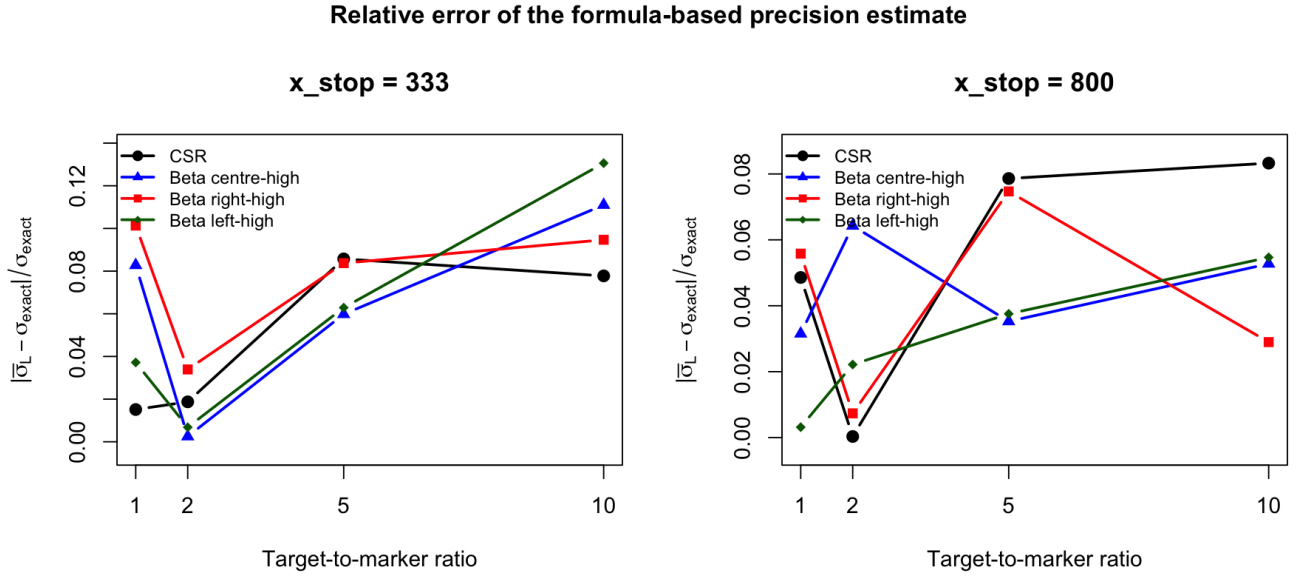


Figure 4: Relative precision gap against the target-to-marker ratio for $x_{\text{stop}} = 333$ (left) and $x_{\text{stop}} = 800$ (right). Colours distinguish the four spatial scenarios.

Both precision measures increased with the target-to-marker ratio. At $x_{\text{stop}} = 333$, $\bar{\sigma}_L$ increased from approximately 8.3% at $\bar{u} = 1$ to approximately 18.6% at $\bar{u} = 10$. σ_{exact} ranged from 8.02% to 9.25% at $\bar{u} = 1$ and from approximately 20% to 22% at $\bar{u} = 10$. At $x_{\text{stop}} = 333$ and $\bar{u} = 10$, the four values of $\bar{\sigma}_L$ ranged from 18.55% to 18.77%.

Across the eight settings at each ratio, the mean relative gaps were 0.047, 0.020, 0.065 and 0.079 for $\bar{u} = 1, 2, 5$ and 10, respectively. The relative gap was smaller at $x_{\text{stop}} = 800$ than at $x_{\text{stop}} = 333$ in 12 of the 16 matched comparisons of spatial scenario and ratio, including all four scenarios at $\bar{u} = 5$. The signed gap was negative in 25 of the 32 settings and in all 16 settings at $\bar{u} = 5$ and $\bar{u} = 10$.

Centre-high had the largest CSR gap, 0.420. Its relative precision gap was the largest in one of the eight combinations of ratio and stopping count.

6.4 Precision gap versus mean marker count

Figure 5 shows the relative precision gap against the mean counted marker number $\bar{n}$ for all 32 settings. The horizontal axis is logarithmic, colours distinguish the four spatial scenarios, and symbols distinguish the two stopping counts.

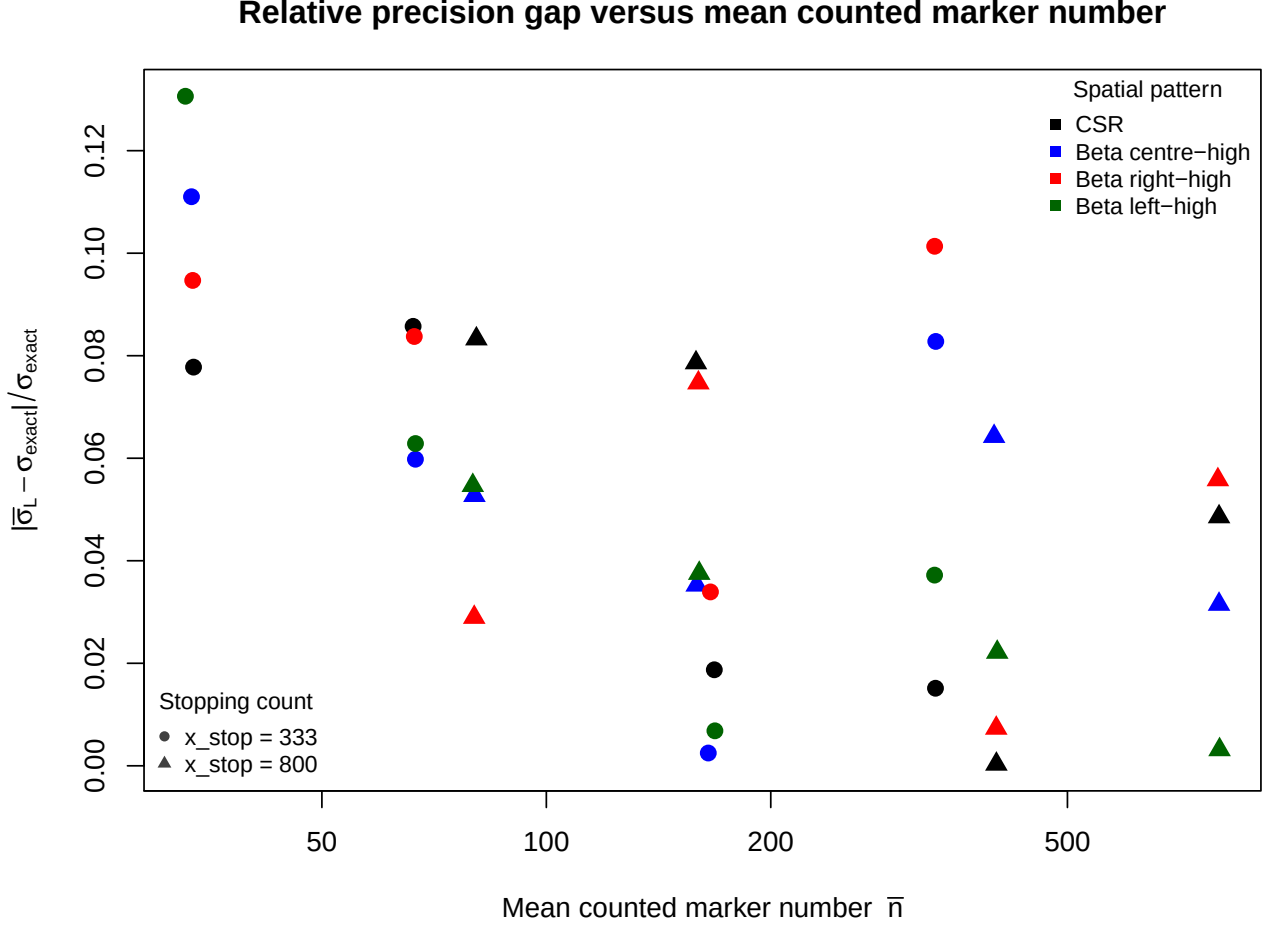


Figure 5: Relative precision gap against the mean counted marker number $\bar{n}$ for all 32 settings. The horizontal axis is logarithmic; colours identify the four spatial scenarios, circles indicate $x_{\text{stop}} = 333$, and triangles indicate $x_{\text{stop}} = 800$.

Across the 32 settings, the relative precision gap tended to be smaller at larger values of $\bar{n}$.

At $\bar{u} = 10$ and $x_{\text{stop}} = 333$, $\bar{n}$ ranged from 32.8 to 33.6 across the four scenarios. At $\bar{u} = 1$ and $x_{\text{stop}} = 800$, it ranged from 795.7 to 799.9.

The largest relative precision gap, 0.131, occurred in the left-high scenario at $\bar{u} = 10$ and $x_{\text{stop}} = 333$, where $\bar{n} = 32.8$. The smallest gap, 0.0003, occurred in the CSR scenario at $\bar{u} = 2$ and $x_{\text{stop}} = 800$, where $\bar{n} = 401.5$.

At $\bar{u} = 1$ and $x_{\text{stop}} = 333$, the range of $\bar{n}$ across the four scenarios was 1.2, from 331.7 to 332.9, while the corresponding relative precision gaps ranged from 0.015 to 0.101.

6.5 Concentration bias

Figure 6 shows the absolute relative bias of the concentration estimate based on the mean marker count, $\hat{c}_{\text{Jensen}}$, against the target-to-marker ratio for the two stopping counts.

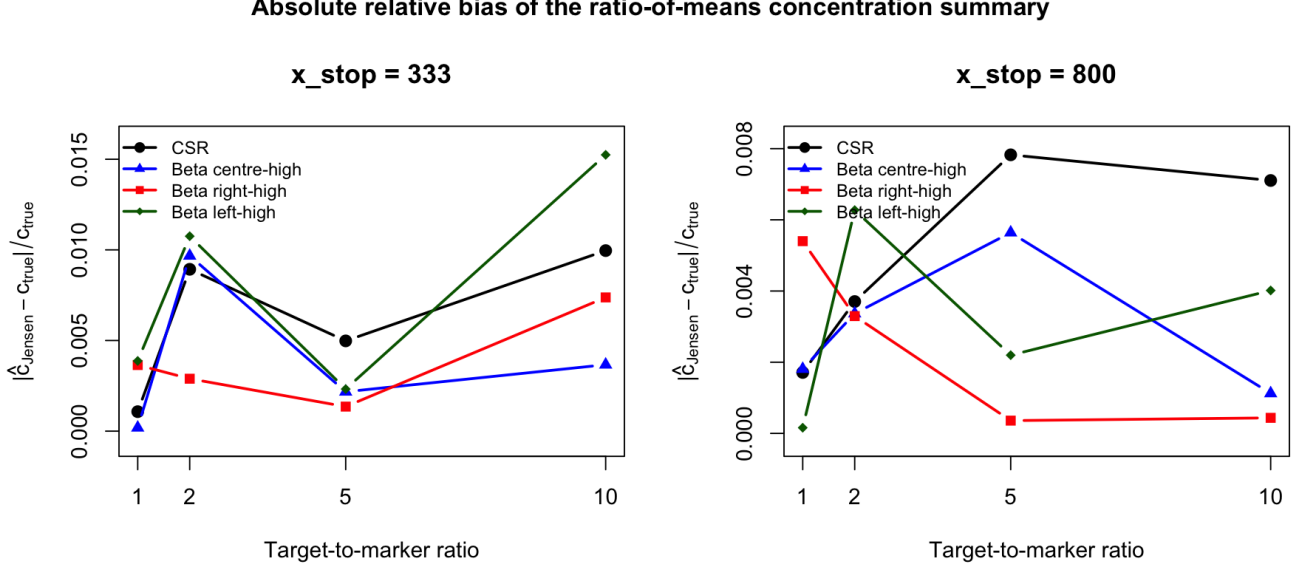


Figure 6: Absolute relative bias of the concentration estimate based on the mean marker count against the target-to-marker ratio for $x_{\text{stop}} = 333$ (left) and $x_{\text{stop}} = 800$ (right). Colours distinguish the four spatial scenarios.

The absolute relative bias of $\hat{c}_{\text{Jensen}}$ ranged from approximately 0.02% to 1.52%, with a mean of 0.45%. The largest value occurred in the left-high scenario at $\bar{u} = 10$ and $x_{\text{stop}} = 333$.

The absolute raw bias reached 5.31% in the left-high scenario at $\bar{u} = 10$ and $x_{\text{stop}} = 333$. Across the remaining 31 settings, its largest value was 3.42%. The bias component accounted for at most 6.1% of σ_{exact}^2 and had a median contribution of 1.4%.

7 Discussion

7.1 Interpretation of the main findings

This project examined whether spatial heterogeneity reduced the reliability of the linear-method precision formula. Within the shared-pattern settings examined, departure from CSR did not consistently reduce agreement between the formula-based estimate and the simulation benchmark. This conclusion is limited to the design used here, in which targets and markers followed the same spatial distribution within each scenario.

The smallest mean relative precision gap occurred at $\bar{u} = 2$ (2.0%). This setting matches the practical balance identified by Mays et al. (2025, 4) and lies at the upper end of the preferred range reported by Price et al. (2016, 78). It is also consistent with Maher’s observation that, at $u = 1$, half of the counted specimens are markers rather than biological targets (1981, 156–57). The present

result therefore provides a useful point of agreement with the earlier practical guidance: within the settings examined, the precision formula agreed most closely with the simulation benchmark at a ratio already favoured on counting-efficiency grounds. It does not establish $\bar{u} = 2$ as a universal optimum, because the earlier recommendations concern precision and counting efficiency, whereas the present comparison concerns agreement between two precision estimates.

A plausible explanation follows from this shared-pattern design. Targets and markers had the same expected spatial pattern, so regions with a higher expected target density also had a higher expected marker density. Because the concentration estimate in Equation (6) depends on the ratio of the two counted numbers, these common spatial changes can partly cancel in that ratio. Spatial heterogeneity can therefore change where the counting rectangle ends and how many markers are counted without necessarily producing a corresponding change in the precision gap. This explanation does not extend to preparations in which targets and markers follow different spatial distributions.

The CSR gap and the precision gap also describe different features of the simulation. The CSR gap summarises the overall spatial pattern of the target points, whereas the precision gap depends on the target and marker counts inside the final counting rectangle. The ordering of the scenarios by CSR gap therefore does not have to match their ordering by precision gap. This does not show that spatial pattern is unimportant. It shows only that the size of the departure from CSR was not, by itself, a consistent indicator of agreement between the two precision estimates in the settings examined.

The larger gaps at lower $\bar{n}$ should not be taken as evidence that the marker count alone caused the disagreement. A smaller n_m increases the $1/n_m$ term in Equation (11), and changes in a small denominator have a greater effect on the concentration estimate. However, $\bar{n}$ changed together with the target-to-marker ratio and the stopping count, so this simulation cannot separate their effects.

7.2 Limitations

The simulation was designed to answer a controlled question, so its findings apply only to the settings examined. The four Beta patterns were chosen to produce clear departures from CSR rather than fitted to real microscope slides. Only four spatial patterns, four target-to-marker ratios and two stopping counts were examined. The target and marker points were generated independently but from the same coordinate distribution in each scenario. This kept their expected spatial relationship the same while the spatial pattern was changed. It also excluded cases in which targets and markers follow different spatial patterns. The results therefore cannot be extended to those cases.

The counting procedure was also simplified. All simulations used the same horizontal transect band, $y \in [0.83, 0.88]$, and the same left-to-right counting direction. Changing the position or direction of the transect could produce different marker counts under a non-uniform pattern. The simulation also generated specimen coordinates directly, so it did not include variation introduced during slide preparation, mixing, specimen overlap or microscopic identification. The results describe the statistical behaviour of the counting method rather than every source of uncertainty in a real preparation.

Because $\bar{n}$ changed with the ratio and stopping count, its association with the precision gap is

descriptive. The CSR diagnostic was based on one representative point pattern for each scenario and did not measure variation between repeated patterns. Finally, the simulation benchmark was calculated from 400 repetitions. It therefore retains some Monte Carlo variation, while the precision gaps were reported as point values without Monte Carlo intervals.

7.3 Implications and future work

For a method that depends on both target and marker counts, the target pattern alone is not enough to judge reliability, although this conclusion currently applies only when targets and markers share the same spatial pattern.

The next simulation step should generate targets and markers from different spatial distributions. This could include a non-uniform target pattern with approximately uniform markers, as well as target and marker patterns with opposing spatial gradients. These settings would test whether agreement with the formula changes when the shared-pattern explanation in Section 7.1 no longer applies. A later extension could introduce a clustered pattern using a Thomas process, which would represent a form of spatial aggregation not included in the present Beta settings (Diggle 2013, 100).

The relationship with $\bar{n}$ could be examined using settings that produce similar mean marker counts from different combinations of target-to-marker ratio and stopping count. The CSR diagnostic should also be repeated across multiple point patterns within each scenario. More Monte Carlo repetitions would allow intervals to be reported for the precision gaps, while different transect positions and counting directions would test the sensitivity of the results to the counting geometry.

In the longer term, validation against real microscope preparations would test how far these simulation-based findings carry over to practical counting.

7.4 Conclusion

This report presents an initial controlled assessment of the linear-method precision formula under non-uniform specimen distributions. The analysis compared formula-based precision with variation across repeated simulations under CSR and three Beta patterns, using four target-to-marker ratios and two stopping counts. In all settings examined, targets and markers followed the same spatial distribution.

Within this shared-pattern design, a stronger departure from CSR was not consistently associated with a larger precision gap. Larger gaps were more often observed when fewer markers were counted, but marker count changed together with the target-to-marker ratio and stopping count. Their separate effects therefore could not be determined from the present design.

These findings should not be interpreted as a general validation of the precision formula under spatial heterogeneity. The present analysis does not cover situations in which targets and markers follow different spatial distributions, which may produce different results. Mays et al. (2025, 28) identified inhomogeneity as an area lacking quantitative assessment; this report provides an initial step towards that assessment rather than a complete answer. The next stage should examine different target and marker distributions before the findings are compared with real microscope preparations.

8 Acknowledgements

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The code and outputs required to reproduce this report are available at https://github.com/SHERRY-DOGGIE-1/MDS_Final_Report.

Anthropic’s Claude was used to assist with language polishing and to check the manuscript for clarity, consistency and referencing errors. All research design, analysis, interpretation and conclusions are the author’s own.

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10 Appendix

10.1 Additional theoretical and practical background

10.1.1 Laboratory-processing uncertainty

The simulations begin after target and marker locations have been generated and therefore do not represent chemical preparation or specimen loss. In experiments using marine sediment samples, Price et al. (2016, 78) measured an average loss of 8.2% of *Lycopodium* spores during processing, corresponding to a 4.8% overestimate of dinoflagellate-cyst concentration. This is an important practical source of uncertainty, but it is outside the scope of the present simulation.

10.1.2 Equal counts under fixed counting effort

The theoretical result at $u = 1$ can be seen from the two counting terms in the linear-method precision formula. For a fixed expected combined count $T = x + n$,

$$\frac{1}{x} + \frac{1}{n} = \frac{T}{xn}.$$

For fixed T , the product xn is largest when $x = n = T/2$. The counting contribution is therefore smallest when the expected target and marker counts are equal. This is a planning result for fixed expected counting effort and does not require the observed counts to be identical in every repetition (Regal and Cushing 1979, Eq. 6.2, p. 564).

10.1.3 Inter-event distance reference under CSR

For two events with coordinates (X_i, Y_i) and (X_j, Y_j) , their inter-event distance is

$$d_{ij} = \sqrt{(X_i - X_j)^2 + (Y_i - Y_j)^2}.$$

The empirical cumulative distribution $\hat{H}(t)$ records the proportion of observed inter-event distances no greater than t . Under CSR on the unit square, the corresponding distribution for $0 \leq t \leq 1$ is

$$H_{\text{CSR}}(t) = \pi t^2 - \frac{8}{3}t^3 + \frac{1}{2}t^4, \quad 0 \leq t \leq 1. \quad (18)$$

This is the unit-square inter-event distance distribution given by Diggle (2013, Eq. 2.2, pp. 18–19).

10.2 Complete formula and simulation precision results

Table 5 reports the formula-based precision estimate, simulation benchmark and precision gaps for all 32 settings.

Table 5: Formula-based and simulation precision results for all 32 settings.

Scenario	$\bar{u}$	x_{stop}	$\bar{n}$	$\bar{\sigma}_L$ (%)	σ_{exact} (%)	Δ	Rel. gap
Beta centre-high	1	333	332.9	8.31	9.06	-0.75	0.083
Beta left-high	1	333	331.7	8.31	8.02	0.30	0.037
Beta right-high	1	333	331.8	8.32	9.25	-0.94	0.101
CSR	1	333	332.6	8.31	8.44	-0.13	0.015
Beta centre-high	2	333	164.9	10.00	9.97	0.02	0.002
Beta left-high	2	333	168.3	9.94	9.87	0.07	0.007
Beta right-high	2	333	166.0	9.98	10.33	-0.35	0.034
CSR	2	333	168.0	9.94	10.13	-0.19	0.019
Beta centre-high	5	333	66.7	13.82	14.70	-0.88	0.060
Beta left-high	5	333	66.8	13.82	14.75	-0.93	0.063
Beta right-high	5	333	66.5	13.85	15.11	-1.27	0.084
CSR	5	333	66.3	13.86	15.16	-1.30	0.086
Beta centre-high	10	333	33.4	18.62	20.95	-2.33	0.111
Beta left-high	10	333	32.8	18.77	21.59	-2.82	0.131
Beta right-high	10	333	33.5	18.57	20.52	-1.94	0.095
CSR	10	333	33.6	18.55	20.12	-1.56	0.078
Beta centre-high	1	800	798.6	5.82	6.01	-0.19	0.032
Beta left-high	1	800	799.9	5.82	5.83	-0.02	0.003
Beta right-high	1	800	795.7	5.82	6.17	-0.34	0.056
CSR	1	800	798.6	5.82	5.55	0.27	0.049
Beta centre-high	2	800	398.7	6.82	6.40	0.41	0.064
Beta left-high	2	800	402.5	6.80	6.95	-0.15	0.022
Beta right-high	2	800	401.3	6.80	6.75	0.05	0.007
CSR	2	800	401.5	6.80	6.80	0.00	0.000
Beta centre-high	5	800	159.1	9.20	9.53	-0.34	0.035
Beta left-high	5	800	160.4	9.17	9.53	-0.36	0.038
Beta right-high	5	800	159.9	9.18	9.92	-0.74	0.075
CSR	5	800	158.8	9.21	9.99	-0.79	0.079
Beta centre-high	10	800	80.1	12.15	12.83	-0.68	0.053
Beta left-high	10	800	79.7	12.18	12.88	-0.70	0.055
Beta right-high	10	800	80.0	12.15	12.51	-0.36	0.029
CSR	10	800	80.6	12.12	13.22	-1.10	0.083